VEAGLE: Eye Gaze-Assisted Guidance for Video Browser Showdown

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Abstract. In this work, we focus on assisting users in finding information that may have been unintentionally overlooked. Our system supports not only experienced users but also newcomers to Video Browser Showdown systems, enabling them to search for information more quickly and accurately. During the querying process, users might unintentionally miss important images, including those they are specifically looking for. By leveraging eye-tracking technology, our system records the user's gaze duration on each image and optimises the presentation of potentially relevant images, which should lead to better performance when using interactive systems under time-pressure.

Keywords: video browser showdown · retrieval systems · eye-tracking · semantic embedding · vector database

1 Introduction

The Video Browser Showdown (VBS) [13–16] is an annual international competition that challenges researchers in the field of video retrieval to develop and demonstrate cutting-edge techniques and systems. Participants compete to effectively search and retrieve relevant video content from a large dataset based on a variety of complex and dynamic queries. The competition provides a platform for showcasing innovative approaches in video analysis, indexing, and retrieval, emphasizing both the precision and speed of the solutions. By fostering collaboration and knowledge exchange among experts, the VBS aims to advance the state-of-the-art in video search technologies and promote the practical application of these advancements in real-world scenarios.

Eye tracking [6] is a technology that measures and records eye movements to determine where and how long a person is looking. By capturing the focus

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and attention of the eyes, eye tracking provides valuable insights into visual behavior and cognitive processes. This technology has a wide range of potential applications, from enhancing user experiences in virtual reality and gaming [3], to improving the design of websites and user interfaces, to advancing medical diagnostics and research in psychology and neuroscience. With its ability to offer precise and real-time data, eye tracking holds significant promise for transforming various fields and creating more intuitive and effective human-computer interactions.

Integrating eye tracking into the retrieval systems is the core focus of our research. By monitoring where and how long users focus their gaze on different elements, eye tracking provides real-time feedback that has the potential to enhance the accuracy and relevance of search results. For instance, an interactive system can record the duration of a user's gaze on various images, identifying those that may have been briefly viewed but could be highly relevant to the search query. By re-prioritizing and suggesting these images, the retrieval system could not only improve search efficiency but also deliver a more personalized and intuitive user experience. This fusion of eye tracking and retrieval systems is at the heart of our system, holding significant potential for advancing search technologies and optimizing user interactions across various applications.

2 Related Work

Existing System: Current systems, in addition to focusing on easy-to-use interfaces for users, also integrate a number of advanced artificial intelligence models to solve Video Browser Showdown tasks. Some common systems are VISIONE 5.0 [1], ViewsInsight [17], PraK Tool [7], *etc.*.

Amato *et al.* [1] present VISIONE 5.0, an advanced system featuring an enhanced user interface and sophisticated AI models designed for the Video Browser Showdown (VBS) 2024. VISIONE 5.0 aims to significantly improve user experience and retrieval performance by incorporating intuitive interaction mechanisms and cutting-edge artificial intelligence techniques. Vuong *et al.* [17] propose ViewsInsight, a novel approach designed to enhance video retrieval for the Video Browser Showdown (VBS) 2024. ViewsInsight focuses on providing a user-friendly interaction mechanism that streamlines the search process and improves retrieval accuracy. By incorporating advanced features such as intuitive search interfaces and real-time feedback, ViewsInsight enables users to quickly locate relevant video content. PraK Tool, an interactive search tool designed to leverage video data services introduced by Lokoc *et al.* [7].

PraK Tool leverages keyframes and features extracted from the VISIONE system while allowing integration of new feature extraction models. It splits functionality into video data services, which manage large datasets, and application services for user interaction. This separation reduces workload and complexity through efficient microservice architecture.

Schall *et al.* [12] focuses on known-item search by utilizing logged query data from previous Video Browser Showdown competitions. They address how the use of a different model for formulating queries would impact the ranking of target videos or shots. By re-evaluating each query with these different extracted embeddings, they calculate the video-rank and shot-rank based on relevance to the specified target videos and time frames. This approach leverages existing data and new models, providing insights into their performance using a large volume of historical data.

Other teams tried to streamline their workflows by enhancing existing modalities such as VERGE or replacing embedding models to improve retrieval performance in free-text and visual similarity searches, as seen with DiveXplore, which optimizes server-client response times.

In our approach, while we strive to gather comprehensive information from users, we recognize that user behaviors contain valuable insights that remain underutilized. To address this, we are collecting eye movement data for analysis, referred to as the Eye-tracking Based System. We aiming to gain a deeper understanding of users' implicit interests identifying what they seek. And discerning what they choose to overlook.

Eye-Tracking Based System: Recent advancements in eye-tracking [6] technology are driven by innovative approaches. Zhao *et al.* [18] demonstrate the use of special cameras that capture rapid scene changes for more accurate and real-time eye-tracking data, essential for applications requiring fast eye movement detection. Klaib *et al.* [5] focus on integrating machine learning [8] for eye tracking, enhancing accuracy and efficiency through various algorithms and techniques. In medicine, Harezlak *et al.* [4] survey the use of eye-tracking technology in advancing medical research and practice, including diagnosing neurological disorders, monitoring cognitive states, and improving surgical training. Majaranta *et al.* [9] examine how eye-tracking enhances Human-Computer Interaction (HCI) [10]. They review eye-based interaction techniques, such as gaze-based pointing, selection, and control, and explore the psychological aspects of gaze behavior to improve HCI system design. The study highlights the creation of more intuitive and natural user interfaces by integrating eye-tracking data with other input modalities, enhancing user experience and accessibility.

In our proposed system, we leverage eye-tracking technology to enhance the state of the art in video search by providing a more personalized and efficient user experience. Research indicates that users often miss relevant images during rapid browsing due to time constraints, leading to potential oversights of critical contentecting eye-tracking data, our system identifies images that users have overlooked, enabling us to re-highlight these crucial visuals in subsequent searches. This targeted approach not only aids in rediscovering important images but also optimizes the viewing experience by removing non-essential images from the viewing list. As a result, users can focus on significant content, reducing cognitive load and eye strain during the search process. This innovative integration of eye-tracking with video retrieval enhances user engagement and ensures that relevant information is not inadvertently missed.

3 System Overview

3.1 Overview

In traditional retrieval systems [11] [2], users interact with the system through a keyboard and mouse, resulting in static search results that require further interaction to refine and filter the necessary information. This process is not only time-consuming but also prone to missing important information when a user is browsing result sets under time-pressure.

In contrast, our system introduces a completely novel approach by integrating an eye-tracking device to measure the user's gaze in real-time. With this feature, the system can automatically re-suggest potentially relevant images that users might have unintentionally overlooked during their browsing. This creates a smoother and more seamless user experience.

Our system's dynamic mechanism is fundamentally different from traditional search systems with static results. Instead of providing a fixed list, our system continuously updates and adjusts the image list based on the user's behavior and gaze. This not only saves users time but also ensures that they do not miss any important information, potentially offering a more efficient and intuitive interaction experience.

3.2 Eye Movement-Based Method

Eye tracking is employed in our system to record eye movements as users observe the screen. When users focus on specific areas where images are displayed, the system measures the duration of their gaze on each image. After users proceed to the next action, the system sorts the images based on the time spent observing them, identifying which images received the least attention. This recording of eye gestures allows us to discern which parts of the screen the users are concentrating on and which parts they might unintentionally overlook.

The use of eye tracking provides valuable insights into user behavior and preferences. By analyzing where and for how long users focus their gaze, we can determine which content captures their interest and which content is disregarded. This data is crucial for optimizing the user interface and enhancing the overall user experience.

Furthermore, understanding viewing patterns can help improve the design of interactive elements. By knowing where users are likely to look first, we can prioritize placing important information (such as the highest scoring image to look for) or interactive buttons strategically in those areas, in order to increase usability and efficiency. This approach can reduce the cognitive load on the user, making the interface more intuitive and easier to navigate.

3.3 System Architecture

The architecture of our system is illustrated in Fig. 1, and the detailed description of the system is provided below.

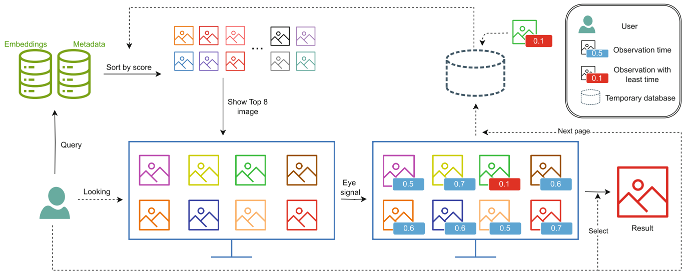

Fig. 1. Architecture of our system

First, the user initiates an query representing their information need. Based on the input received from the user, our system queries the Embeddings Database and Metadata to select the most relevant results. These results are then ranked in descending order according to their scores, meaning that the most accurate result appears first.

On the user interface, following the ranked order, the top eight images are displayed for the user to observe and select from. At this point, the Eye Tracking system becomes active, recording the user's eye movements to determine which parts of the screen they are focusing on and which parts they may have overlooked.

The system then has two possible courses of action: If the user finds the desired image, the search process concludes. However, if the user does not find the desired image, they can interact with a button on the screen to load the next set of images. At this stage, based on the previously recorded observation times, we establish an α threshold for image accuracy and a β threshold for observation time (as formula 1)[1]. If an image that was overlooked has a score higher than alpha and an observation time less than beta, it is flagged for reconsideration and re-ranked for display in the next iteration. This process repeats until the user finds the desired image.

$$\begin{cases} Image_i(Score) \geq \alpha \\ Image_i(TimeLooking) \leq \beta \end{cases} \tag{1}$$

Expanding on this, when a user inputs a query, our system processes it using an algorithm that integrates two databases: a vector database (Milvus) and a Metadata database (Elasticsearch). Milvus stores semantic vectors and enables efficient retrieval through approximate nearest neighbor searches, while Elasticsearch manages metadata and and text matching.

The user interface displaying the top eight images in a user-friendly manner, it allows users to quickly assess and choose from the most relevant options. The

[1] Scores are normalized to 0–1, with α and β thresholds set at 0.5. α reflects accuracy (0–100%), while β scales observation time from 0 up to 10 s or the maximum viewed time.

seamless integration of the Eye Tracking system provides an additional layer of interactivity. By capturing and analyzing the user's gaze in real-time, the system gains valuable insights into user behavior and preferences. As illustrated in Fig. 2 our user interface primarily displays images, without any buttons or input fields that might distract the user, allowing them to focus on viewing and interacting with the content.

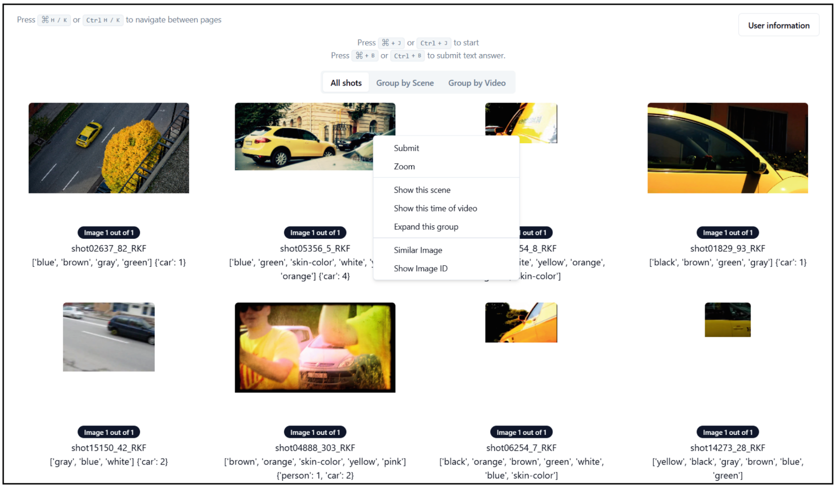

Fig. 2. A screenshot of the system user interface

This real-time feedback mechanism is crucial. If the user does not find the desired image within the initial set, the system intelligently adapts by using the recorded eye movement data. By setting the alpha and beta thresholds, the system can dynamically adjust the ranking of images (on the next screen), ensuring that potentially relevant images that were initially overlooked are given another chance to be reviewed by the user.

This iterative process continues, making the system highly adaptive and responsive to user needs. By continuously refining the search results based on user interactions, the system aims to minimize the time and effort required for users to find the desired images.

4 Conclusion

In conclusion, the application of eye tracking technology in our system effectively addresses the challenges posed by the Video Browser Showdown. By capturing and analyzing real-time user gaze data, the system can dynamically adjust search results and prioritize images based on user attention and interaction patterns. The performance of this system will be evaluated in competition at VBS 2025.

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